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Problem Statement:

A Gym operator would like to analyse the fitness patterns and performance across diverse gym experience levels. The dataset provides a detailed overview of gym members' exercise routines, physical attributes, and fitness metrics. It contains 4000 over samples of gym data, including key performance indicators such as heart rate, calories burned, and workout duration. Each entry also includes demographic data and experience levels, allowing for comprehensive analysis of fitness patterns, athlete progression, and health trends.

To start off this assignment, I will follow the CRISP-DM framework to complete. Starting with the first phase, **business understanding**, which is to understand and improve gym member outcomes by analysing the relationships between experience levels, exercise routines, and physical performance and offer insights into how different factors affect the target variable “Calories_Burned”.

Once I read and understood the project objectives, I continued with the next phase, **data understanding.** I imported the csv file provided into KNIME and used the csv reader node which allows me to view the data in table format and the statistics format.

Rows: 4865 | Columns: 15

ge	Gender	Weight (k...	Height (m)	Max_BPM	Avg_BPM	Resting_...	Session_...	Calories_...	Workout_...	Fat_Perc...	Water_Int...	Workout_...	Experien...	BMI
6	Male	88.3	1.71	180	157	60	1.69	1313	Yoga	12.6	3.5	4	3	30.2
6	Female	74.9	1.53	179	151	66	1.3	883	HIIT	33.9	2.1	4	2	32
2	Female	68.1	1.66	167	122	54	1.11	677	Cardio	33.4	2.3	4	2	24.71
5	Male	53.2	1.7	190	164	56	0.59	532	Strength	28.8	2.1	3	1	18.41
8	Male	46.1	1.79	188	158	68	0.64	556	Strength	29.2	2.8	3	1	14.39
6	Female	58	1.68	168	156	74	1.59	1116	HIIT	15.5	2.7	5	3	20.55
6	Male	70.3	1.72	174	169	73	1.49	1385	Cardio	21.3	2.3	3	2	23.76
0	Female	69.7	1.51	189	141	64	1.27	895	Cardio	30.6	1.9	3	2	30.57
8	Male	121.7	1.94	185	127	52	1.03	719	Strength	28.9	2.6	4	2	32.34
8	Male	101.8	1.84	169	136	64	1.08	808	Cardio	29.7	2.7	3	1	30.07
1	Male	120.8	1.67	188	146	54	0.82	593	HIIT	20.5	3	2	1	43.31
3	Male	51.7	1.7	175	152	72	1.15	865	HIIT	23.6	3.5	3	2	17.89
7	Male	112.5	1.61	195	165	61	1.24	1013	Cardio	22.1	2.7	3	2	43.4
1	Male	94.5	2	179	136	69	1.18	794	HIIT	27.6	3.7	3	1	23.62
0	Male	117.7	1.81	196	161	54	1.35	1195	Yoga	26.4	3.3	3	2	35.93
9	Female	42.5	1.75	181	131	52	1.13	740	Strength	26.2	2.1	2	1	13.88
9	Female	64	1.53	166	167	58	1.33	1111	HIIT	29.8	2.3	3	2	27.34
1	Female	43.8	1.77	182	165	58	1.19	884	Cardio	31.9	1.6	3	1	13.98
7	Female	66.8	1.75	199	146	56	1.13	742	Strength	32.8	2.5	3	2	21.81
5	Female	75.2	1.67	188	167	51	1.37	1030	HIIT	25.2	2.2	2	1	26.96
9	Male	89	1.77	175	127	72	1.5	1048	Strength	28.9	3.7	4	2	28.41
8	Male	71.9	1.77	197	142	72	1.12	875	Cardio	25.7	3.1	2	1	22.95
0	Female	71	1.68	187	161	70	1.17	848	Yoga	33.1	2.5	2	1	25.16
9	Male	120.9	1.78	197	168	65	0.78	721	Yoga	28.1	3.4	2	1	38.16
9	Female	64.3	1.69	190	148	58	1.25	925	HIIT	26.9	1.9	4	2	22.51
2	Female	63.7	1.71	173	169	62	1.42	1080	Yoga	26.1	1.8	3	2	21.78
4	Male	65.2	1.8	192	139	68	0.73	502	HIIT	27.2	2.7	2	1	20.12
9	Male	53.9	1.75	168	135	69	1.48	989	Strength	21.6	2.8	4	2	17.6
5	Male	84.9	1.86	186	136	66	1.64	1104	HIIT	14.2	3.5	5	3	24.54
3	Female	78	1.68	183	135	55	1.29	871	HIIT	32.3	1.8	3	2	27.64
2	Male	108.2	1.8	172	138	53	1.27	964	HIIT	28.4	3.5	3	1	33.4
0	Female	65.4	1.52	185	127	50	1.03	654	Yoga	28	2.2	4	2	28.31
4	Female	50.2	1.61	188	157	67	1.48	1046	HIIT	28.2	2.7	2	1	19.37
4	Female	58.9	1.51	187	157	68	1.04	816	Cardio	31.7	2.5	2	1	25.83

Rows: 13 | Columns: 16

#	RowID	Column	Min	Max	Mean	Std. devi...	Variance	Skewness	Kurtosis	Overall s...	No. missi...	No. NaNs	No. +oos	No. -oos	Median
1	Age	Age	18	59	38.683	12.176	148.253	-0.078	-1.215	188,195	0	0	0	0	
2	Weight	Weight (kg)	40	129.9	73.855	21.199	449.388	0.771	-0.029	359,303	0	0	0	0	
3	Height	Height (m)	1.5	2	1.723	0.128	0.016	0.338	-0.726	8,380.35	0	0	0	0	
4	Max_B	Max_BPM	160	199	179.884	11.521	132.732	-0.038	-1.188	875,135	0	0	0	0	
5	Avg_Bf	Avg_BPM	120	169	143.767	14.339	205.613	0.086	-1.199	699,425	0	0	0	0	
6	Restin(	Resting_BPM	50	74	62.223	7.324	53.642	-0.072	-1.182	302,715	0	0	0	0	
7	Sessio	Session_Duratic	0.5	2	1.256	0.343	0.118	0.026	-0.354	6,112.5	0	0	0	0	
8	Calorie	Calories_Burnec	303	1,783	905.422	272.529	74,272.267	0.278	-0.061	4,404,880	0	0	0	0	
9	Fat_Pe	Fat_Percentage	10	35	24.977	6.257	39.148	-0.634	-0.343	121,512	0	0	0	0	
10	Water_	Water_Intake (l	1.5	3.7	2.627	0.6	0.36	0.071	-1.021	12,778.5	0	0	0	0	
11	Workot	Workout_Frequ	2	5	3.322	0.913	0.833	0.15	-0.808	16,160	0	0	0	0	
12	Experie	Experience_Lev	1	3	1.81	0.739	0.547	0.318	-1.123	8,805	0	0	0	0	
13	BMI	BMI	12.32	49.84	24.912	6.658	44.331	0.763	0.735	121,197.5	0	0	0	0	

Here’s the image of the data in the statistical view, I’ve purposely excluded Gender and Workout_type out as both the columns are string datatypes and wouldn’t contribute statistically.

Rows: 15 | Columns: 14

Name	Type	# Missing val...
Age	Number (Intege	0
Gender	String	0
Weight (kg)	Number (Float)	0
Height (m)	Number (Float)	0
Max_BPM	Number (Intege	0
Avg_BPM	Number (Intege	0
Resting_BPM	Number (Intege	0
Session_Durati	Number (Float)	0
Calories_Burne	Number (Intege	0
Workout_Type	String	0
Fat_Percentage	Number (Float)	0
Water_Intake (li	Number (Float)	0
Workout_Frequ	Number (Intege	0
Experience_Lev	Number (Intege	0
BMI	Number (Float)	0

While doing so, I also managed to find out that this dataset has no missing values in all the columns which makes it easier so I wouldn't need to use a missing value node to remove them.

The next phase in the CRISP-DM framework would be **data preparation.** The first step was to remove duplicates in the dataset.

Rows: 973 | Columns: 15

#	RowID	Age	Gender	Weight (k...	Height (m)	Max_BPM	Avg_BPM	Resting_...	Session_...	Calories_...	Workout_...	Fat_Perc...	Water_Int...	Workout_...	Experi
1	Row0	56	Male	88.3	1.71	180	157	60	1.69	1313	Yoga	12.6	3.5	4	3
2	Row1	46	Female	74.9	1.53	179	151	66	1.3	883	HIIT	33.9	2.1	4	2
3	Row2	32	Female	68.1	1.66	167	122	54	1.11	677	Cardio	33.4	2.3	4	2
4	Row3	25	Male	53.2	1.7	190	164	56	0.59	532	Strength	28.8	2.1	3	1
5	Row4	38	Male	46.1	1.79	188	158	68	0.64	556	Strength	29.2	2.8	3	1
6	Row5	56	Female	58	1.68	168	156	74	1.59	1116	HIIT	15.5	2.7	5	3
7	Row6	36	Male	70.3	1.72	174	169	73	1.49	1385	Cardio	21.3	2.3	3	2
8	Row7	40	Female	69.7	1.51	189	141	64	1.27	895	Cardio	30.6	1.9	3	2
9	Row8	28	Male	121.7	1.94	185	127	52	1.03	719	Strength	28.9	2.6	4	2
10	Row9	28	Male	101.8	1.84	169	136	64	1.08	808	Cardio	29.7	2.7	3	1
11	Row10	41	Male	120.8	1.67	188	146	54	0.82	593	HIIT	20.5	3	2	1
12	Row11	53	Male	51.7	1.7	175	152	72	1.15	865	HIIT	23.6	3.5	3	2

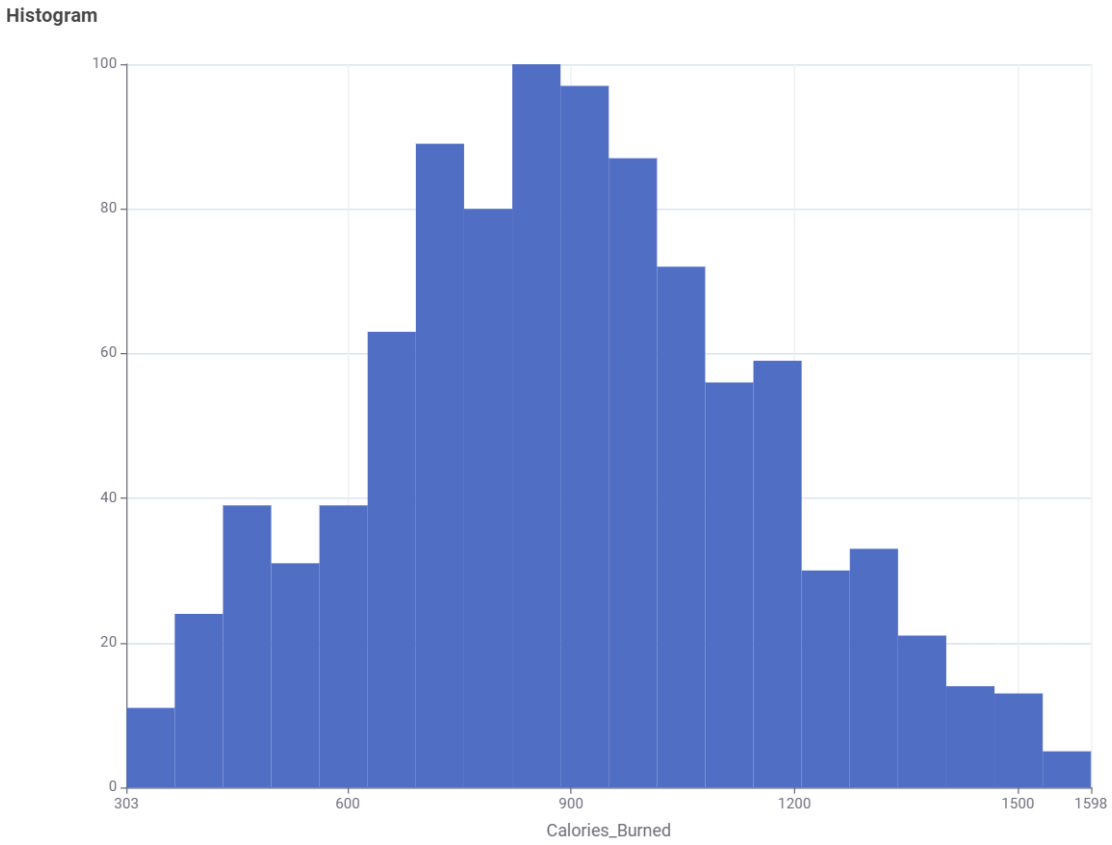
Duplicate analysis revealed the dataset's 4,865 rows consist of only 973 unique records, each repeated exactly 5 times (973 × 5 = 4,865), indicating artificial expansion rather than organic duplication. Duplicates were removed to prevent the model training on redundant information. I also checked each column’s data type, ensuring that it has the correct data type for each column.

Outliers were also identified in BMI (25), Calories_Burned (10), and Weight (8) using the standard 1.5×IQR rule. These observations were kept but replaced with missing values because they represent plausible measurements of gym members rather than obvious data entry errors. Keeping these values ensures the predictive model reflects realistic variations within the population. SAS Viya would also exclude the 10 rows with missing Calories_Burned (the target variable) from training, resulting in an effective sample of 963 rows. Rows with missing predictor values (BMI, Weight) were also kept, as SAS Viya's modeling nodes handle missing inputs internally.

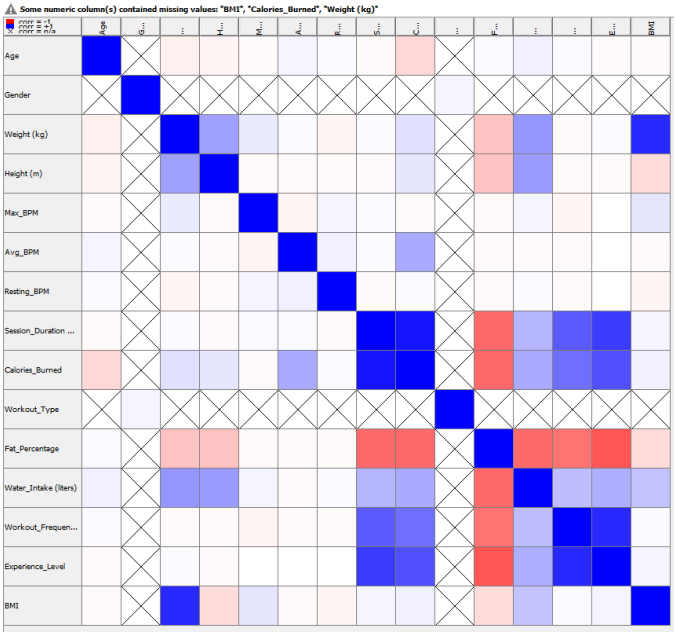
Categorical variables (Gender, Workout_Type, Experience_Level) were left unencoded, as SAS Viya's modeling nodes natively handle nominal inputs; manual encoding would add redundant columns without benefit. Feature engineering was avoided, derived features such as Calories per Hour would introduce data leakage by incorporating the target variable, misaligned with the business objective of explaining (not just predicting) Calories_Burned.

Next I did some exploratory data analysis with our target variable, calories_burned.

Here is a histogram graph of calories burned among the 973 rows of data.

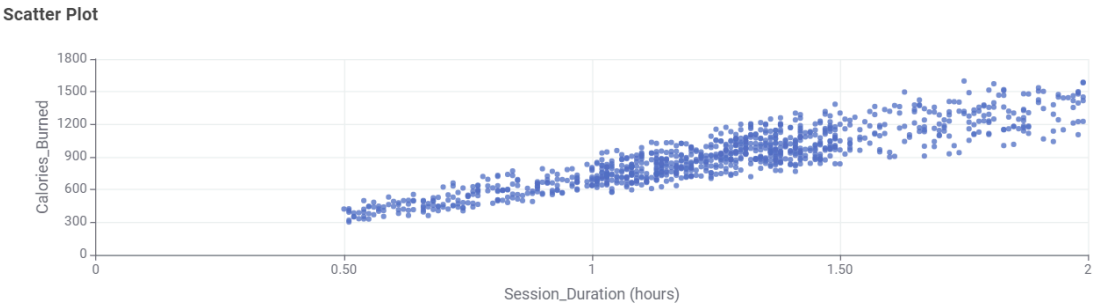


Next I used a linear correlation node to screen which variables highly correlate with calories burned. Next is the correlation heatmap of all the variables with calories burned. As you can see at the calories burned row, there are 3 dark coloured cells. Two blue and one red which are Session_Duration 0.91, Experience_Level 0.69, Fat_Percentage −0.60.

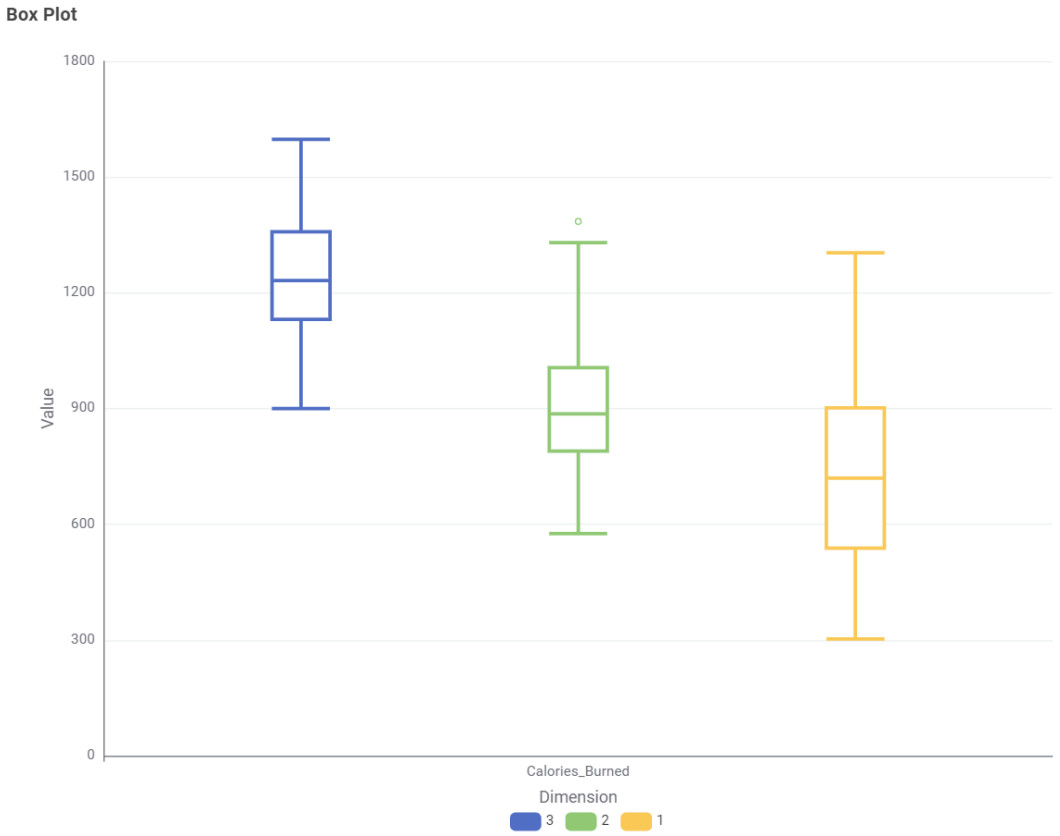


After that, I compared calories burned across the variables that are highly correlated with calories burned, starting with session duration, the strongest relationship with calories burned.

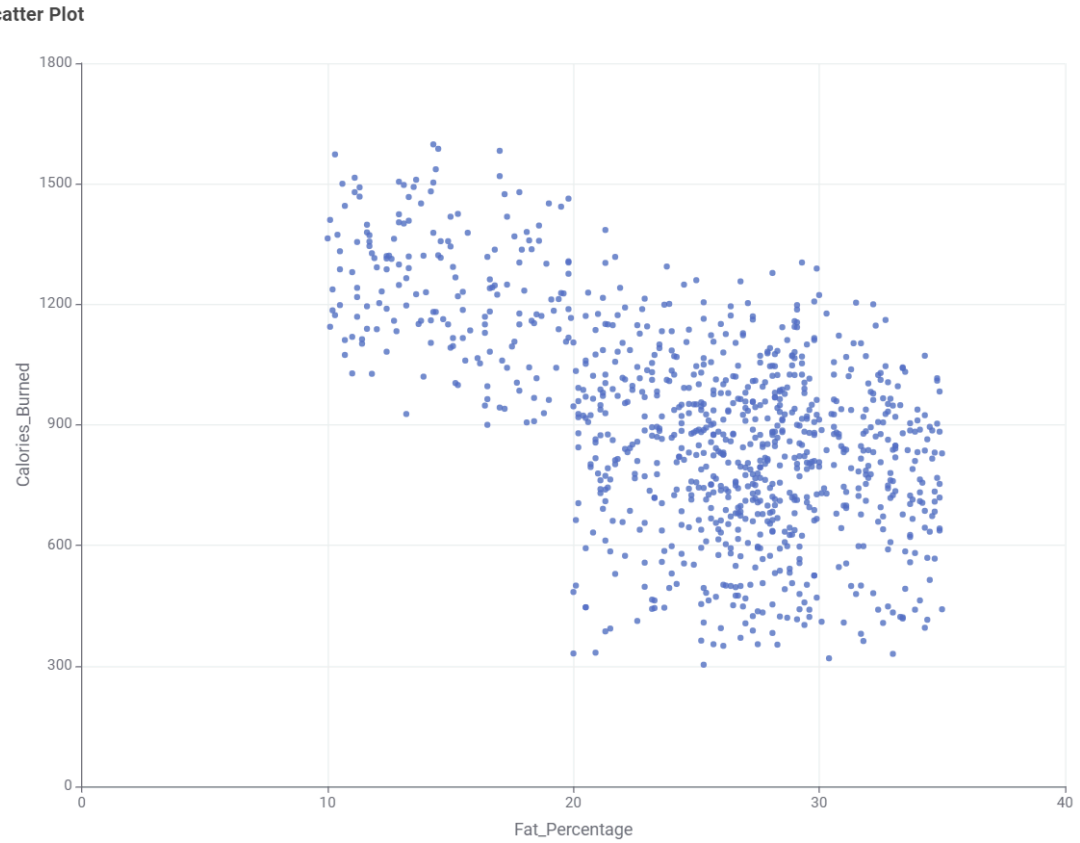
A scatter plot of session duration across calories burned



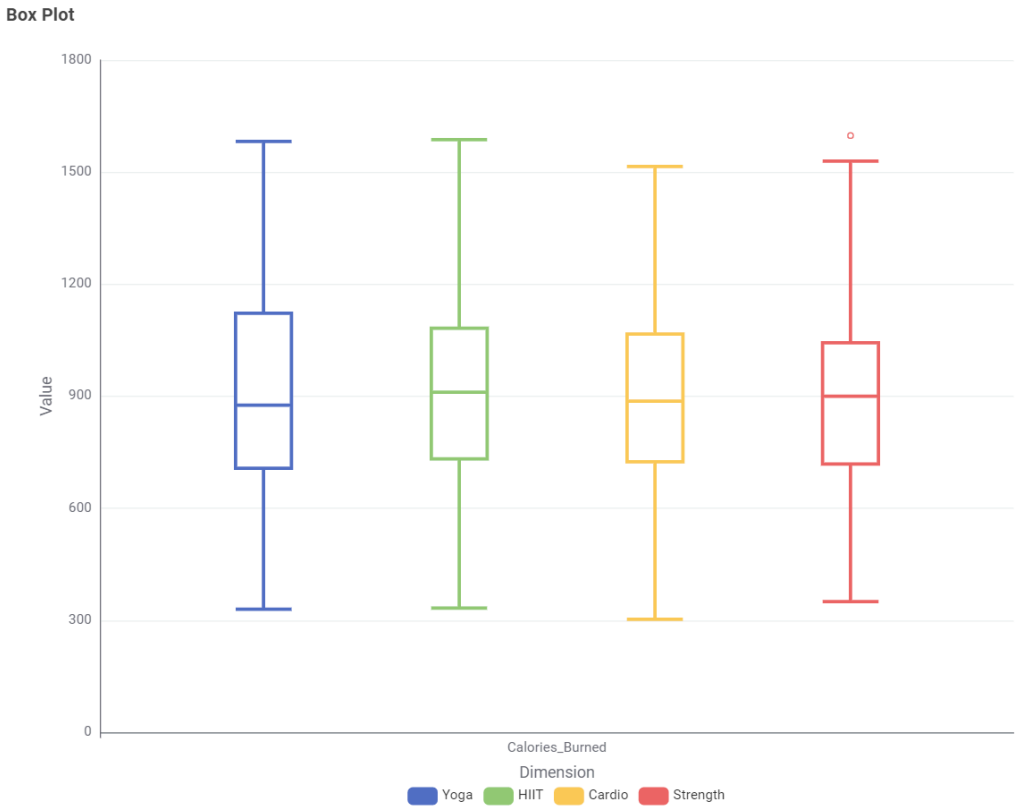
A box plot of experience level across calories burned



Another scatter plot of fat percentage across calories burned



Lastly, this box plot is to find whether the median and spread of calories burned changes across Cardio, HIIT, Strength, and Yoga.



Once data preparation was completed, I exported the cleaned dataset and headed over to SAS viya to move on to the next phase in the CRISP-DM framework, **modelling**. I imported the data into SAS viya and started setting calories burned as the target variable. Next was building the model. Before that, I have to partition the data first. I chose a 70/30 train-validation split to balance model training capacity against validation reliability, given the dataset's moderate size (973 rows). This split retains sufficient training data (~681 rows) for the models, while preserving an adequately sized validation set (~292 rows) to produce stable performance estimates.

Next we move on to the pipelines. I added a Linear regression model node to the data then left the settings as is for now. I ran the pipeline and it was successful. Next, I viewed the results and this was what I found.

Linear Regression

Parameter Estimates

Effect	Parameter	t Value	Sign	Estimate	Absolute Esti...	Standard Error	Pr > |t|	Variance Infla...	Degrees of Fr...
Session_Duration (hours)	Session_Duration (hours)	160.8870	+	709.7565	709.7565	4.4115	0	1.0006	1
Avg_BPM	Avg_BPM	59.4719	+	6.1378	6.1378	0.1032	0.0000	1.0041	1
Intercept	Intercept	42.7702	-	-700.6613	700.6613	16.3820	0.0000	0	1
Gender	Gender Female	28.6333	-	-83.8016	83.8016	2.9267	0.0000	1.0046	1
Age	Age	27.3129	-	-3.3280	3.3280	0.1218	0.0000	1.0068	1

Regression Fit Statistics

Statistic	Description	Value
RMSE	Root MSE	37.3291
RSQUARE	R-Square	0.9795
ADJRSQ	Adj R-Sq	0.9794
AIC	AIC	5,395.6448
AICC	AICC	5,395.7746
SBC	SBC	4,762.0603
TRAIN_ASE	ASE (Train)	1,382.8097
VAL_ASE	ASE (Validate)	1,474.2171

These two images show the coefficient table and fit statistics of the regression model. The Linear Regression model used Calories_Burned as the target, with Session_Duration, Avg_BPM, Age, Gender, and the remaining variables as inputs. The top table shows the parameter estimates. Coefficient signs match the EDA correlations: Session_Duration (β = 709.76) and Avg_BPM (β = 6.14) are positive, consistent with r = 0.91 and r = 0.34; Age (β = −3.33) is negative, consistent with r = −0.15. This agreement between the univariate correlations and the multivariate model supports the reliability of these relationships.

Since coefficient size depends on each variable's scale, t-values were used to rank importance instead. Session_Duration had the highest t-value (160.89), followed by Avg_BPM (59.47), Gender (28.63), and Age (27.31). Session_Duration is clearly the dominant predictor, matching its top ranking in the EDA correlations.

Next we move on to the decision tree. I added a decision tree node from the data source and also left the settings as is for now. Here are the results, with variable importance and fit statistics.

Decision Tree (depth 10, leaf size 5)

Variable Name	Training Importance	Training Relative Importance	Validation Relative Importa...	Count	Validation Importance
Session_Duration (hours)	37,176,179.9201	1	1	32	16,799,240.8588
Avg_BPM	4,495,664.2730	0.1209	0.1236	23	2,076,970.7361
Age	1,436,763.6864	0.0386	0.0374	26	627,927.3138
Gender	917,763.8177	0.0247	0.0172	10	289,307.6323
Fat_Percentage	28,803.1364	0.0008	0.0015	3	25,073.5703
Water_Intake (liters)	45,448.6992	0.0012	0.0013	3	22,231.3808
BMI	4,243.6000	0.0001	-0.0003	1	-5,442.5200
Resting_BPM	44,017.1418	0.0012	-0.0004	4	-7,547.9281
Workout_Type	30,369.7024	0.0008	-0.0012	2	-20,632.1412
Max_BPM	19,489.0788	0.0005	-0.0013	2	-22,385.1764
Weight (kg)	202,080.5206	0.0054	-0.0061	3	-102,907.1096

Target N...	Data Role	Partition ...	Formatt...	Number ...	Average ...	Divisor f...	Root Ave...	Mean Ab...	Root Me...	Mean Sq...	Root Me...	R-Square
Calories_Burned	VALIDATE	0	0	289	4,637.2296	289	68.0972	52.7186	7.2608	0.0060	0.0777	0.9353
Calories_Burned	TRAIN	1	1	674	1,312.5562	674	36.2292	27.6193	5.2554	0.0017	0.0416	0.9804

While Decision Tree achieved a marginally higher training R^2 (0.9804) than Linear Regression (0.9795), it showed clear signs of overfitting: validation RMSE increased from 36.23 to 68.10, nearly double the training error, compared to Linear Regression's much smaller train-to-validation gap (37.19 to 38.40). This is corroborated by several low-importance variables (BMI, Resting_BPM, Workout_Type, Max_BPM, Weight) showing negative validation importance, indicating the splits based on these variables failed to generalize beyond the training data.

We'll move on to the next model, gradient boosting.

Gradient Boosting

Variable Label	Role	Variable Name	Training Importance	Importance Standard Deviat...	Relative Importance
	INPUT	Session_Duration (hours)	983,661.9174	2,826,246.9649	1
	INPUT	Avg_BPM	131,597.5411	296,019.6401	0.1338
	INPUT	Age	34,363.6932	60,550.5310	0.0349
	INPUT	Experience_Level	31,043.0675	233,898.5467	0.0316
	INPUT	Gender	21,496.9677	49,352.0862	0.0219
	INPUT	Fat_Percentage	14,912.7661	127,646.2257	0.0152
	INPUT	Water_Intake (liters)	5,376.8683	25,436.1266	0.0055
	INPUT	Height (m)	4,410.9965	20,277.7578	0.0045
	INPUT	Weight (kg)	4,372.9286	18,178.3830	0.0044
	INPUT	BMI	711.0612	1,547.6106	0.0007
	INPUT	Workout_Type	659.1092	2,246.8117	0.0007
	INPUT	Max_BPM	505.1700	1,236.0303	0.0005
	INPUT	Resting_BPM	329.2821	769.3667	0.0003
	INPUT	Workout_Frequency (days/week)	154.4553	583.0470	0.0002

Target N...	Data Role	Partition ...	Formatt...	Number ...	Average ...	Divisor f...	Root Ave...	Mean Ab...	Root Me...	Mean Sq...	Root Me...	R-Square
Calories_Burned	VALIDATE	0	0	289	498.2033	289	22.3205	16.3177	4.0395	0.0007	0.0268	0.9931
Calories_Burned	TRAIN	1	1	674	101.1642	674	10.0580	7.9510	2.8197	0.0002	0.0125	0.9985

This is the best model out of the three. Gradient Boosting achieved the strongest overall performance (validation RMSE = 22.32, validation R^2 = 0.9931), outperforming both Linear Regression and Decision Tree on unseen data. While its train-to-validation RMSE ratio (2.22x) is proportionally larger than Decision Tree's (1.88x), reflecting boosting's tendency to fit training residuals closely, its absolute validation performance remains substantially better than either alternative. This suggests Gradient Boosting is the most suitable model for this dataset, as its capacity to fit complex patterns outweighs the modest overfitting tradeoff.

However, I decided to go back to the decision tree's settings and see if I could modify it so that it's not overfitting. I've set the max depth from 10 to 6 and the leaf size from 5 to 15. The results solved the overfitting issue but with a trade off. Its gap ratio to the linear regression's 1.03x was from 1.88x to 1.23x. It did decrease but the RMSEs went up. Training RMSE rose from 36.23 to 66.24, and validation RMSE rose from 68.10 to 81.77. R^2 dropped too. Hence, the result is not overfitting but instead underfitting. So i've decided to try one more time with max depth of 8 and leaf size of 10 and this was what returned.

Decision Tree (depth 8, leaf size 10)

Variable Name	Training Importance	Training Relative Importance	Validation Relative Importa...	Count	Validation Importance
Session_Duration (hours)	36,695,331.7944	1	1	13	16,524,269.2027
Avg_BPM	4,419,778.7482	0.1204	0.1220	13	2,016,667.8374
Age	1,027,330.1033	0.0280	0.0225	11	371,357.3399
Gender	803,564.3043	0.0219	0.0189	6	312,226.0785
Water_Intake (liters)	27,084.8000	0.0007	-0.0001	1	-1,678.0800
Weight (kg)	114,481.6641	0.0031	-0.0022	1	-36,861.3318

Target N...	Data Role	Partition ...	Formatt...	Number ...	Average ...	Divisor f...	Root Ave...	Mean Ab...	Root Me...	Mean Sq...	Root Me...	R-Square
Calories_Burned	VALIDATE	0	0	289	6,352.9928	289	79.7057	62.1155	7.8813	0.0138	0.1175	0.9114
Calories_Burned	TRAIN	1	1	674	3,261.0015	674	57.1052	43.6563	6.6073	0.0077	0.0878	0.9514

Decision Tree settings were tuned across three configurations to balance overfitting against predictive accuracy. The default configuration (depth=10, leaf=5) overfit substantially (train-validation RMSE ratio of 1.88x). An initial correction (depth=6, leaf=15) reduced the gap to 1.23x but sacrificed too much accuracy, increasing both training and validation RMSE. A final configuration (depth=8, leaf=10) achieved a better balance, improving both training RMSE (57.11) and validation RMSE (79.71) relative to the depth=6 attempt while maintaining a moderate, acceptable overfitting gap (1.40x). Despite this tuning, Decision Tree's validation RMSE (79.71) remained substantially higher than both Linear Regression (38.40) and Gradient Boosting (22.32), indicating this dataset's relationship with Calories_Burned is better captured by linear or ensemble methods than a single decision tree.

Moving on, it's the **evaluation** phase of the CRISP-DM framework next. Since this is a regression problem, I've set the interval selection statistics from “use rule from project settings” to “Average squared error” then I ran it and this was what returned.

Champion	Name	Algorithm Name	Average Square...	Data Role	Number of Obse...	Root Average S...	Root Mean Abso...	Root Mean Squa...
*	Gradient Boosting	Gradient Boosting	498.2033	VALIDATE	289	22.3205	4.0395	0.0268
	Linear Regression	Linear Regression	1,462.6773	VALIDATE	289	38.2450	5.3944	0.0539
	Decision Tree	Decision Tree	6,352.9928	VALIDATE	289	79.7057	7.8813	0.1175

The Model Comparison node confirmed Gradient Boosting as the champion model, selected automatically based on the lowest Average Squared Error(ASE) on the validation partition (498.20, equivalent to RMSE = 22.32), consistent with the manual model-by-model analysis conducted earlier. Linear Regression (ASE = 1,462.68) and Decision Tree (ASE = 6,352.99) both showed substantially higher validation error. SAS Viya's automated insights identified Session_Duration, Avg_BPM, Age, Experience_Level, and Gender as the five most important predictors in the champion model — Session_Duration and Experience_Level's presence in this list is consistent with their high rankings in the exploratory correlation analysis (r = 0.91 and r = 0.69 respectively), while Age and Gender's importance, despite weaker or absent simple correlation, suggests these variables contribute meaningfully once considered alongside other factors. Now that evaluation is done, we’ll move on to recommendations for the gym operator.

Recommendations

1. Session duration is the single biggest lever available — design programs around it. Session_Duration was the dominant predictor across every method used (r = 0.91 in EDA,

highest t-value in Linear Regression, top variable importance in all three models). The gym operator should treat workout length as the primary driver of calorie outcomes and structure program offerings accordingly — e.g., promoting longer session formats or structured extensions (60 → 90 min classes) as the clearest way to help members hit calorie-burn goals, rather than focusing marketing around workout type alone.

2. Intensity (Avg_BPM) matters, but far less than duration — position it as a secondary lever. Avg_BPM was consistently the second-most important factor (r = 0.34, second-highest t-value, second-ranked in all tree-based importance tables). This supports a "duration first, intensity second" framing for trainers: a longer, moderate-intensity session likely outperforms a short, high-intensity one for total calorie burn. Worth testing this explicitly with trainers if the gym wants to validate programming choices.

3. Experience level should inform program tiering, not just be a background demographic. Experience_Level ranked highly in EDA (r = 0.69) and appeared in Gradient Boosting's top five factors. More experienced members likely sustain longer/higher-output sessions, so the operator could use experience level as a segmentation variable for program design — e.g., structuring progressive session-length increases as beginners advance, rather than treating all experience tiers identically.

4. Age shows a negative relationship worth accounting for in program design. Age had a consistent negative effect (r = −0.15, negative coefficient in Linear Regression, present in Gradient Boosting's top five). This doesn't mean older members should be discouraged — it suggests programs for older demographics may benefit from being designed around session length and consistency rather than intensity, since intensity naturally has physiological limits that shift with age.

5. Gender differences exist and should inform personalized goal-setting. Gender showed a statistically significant effect in Linear Regression (coefficient −83.80, high t-value) and ranked in Gradient Boosting's top five. Rather than a one-size-fits-all calorie target, the operator could use gender-aware baseline expectations when setting member goals, to avoid over- or under-estimating expected outcomes.

6. Workout type, BPM extremes (Max/Resting), Weight, Height, and BMI showed minimal independent predictive value. Despite being intuitive factors, these consistently ranked low across all three models. This is a useful finding in itself — it suggests the gym's messaging around "which workout type burns the most calories" may be less impactful than messaging around "how long and how consistently you train," which is a more actionable and controllable factor for members.

These recommendations are derived from patterns identified in the provided dataset. As noted in the Data Understanding phase, the dataset's structure (973 unique records duplicated exactly fivefold) and the unusually high model fit (R^2 approaching 0.98–0.99) suggest Calories_Burned may follow a defined generative relationship with the input features rather than reflecting fully organic, real-world measurement variability. While the relative importance and direction of these factors remain informative, the gym operator should validate these findings against real member data before making significant operational decisions.